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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

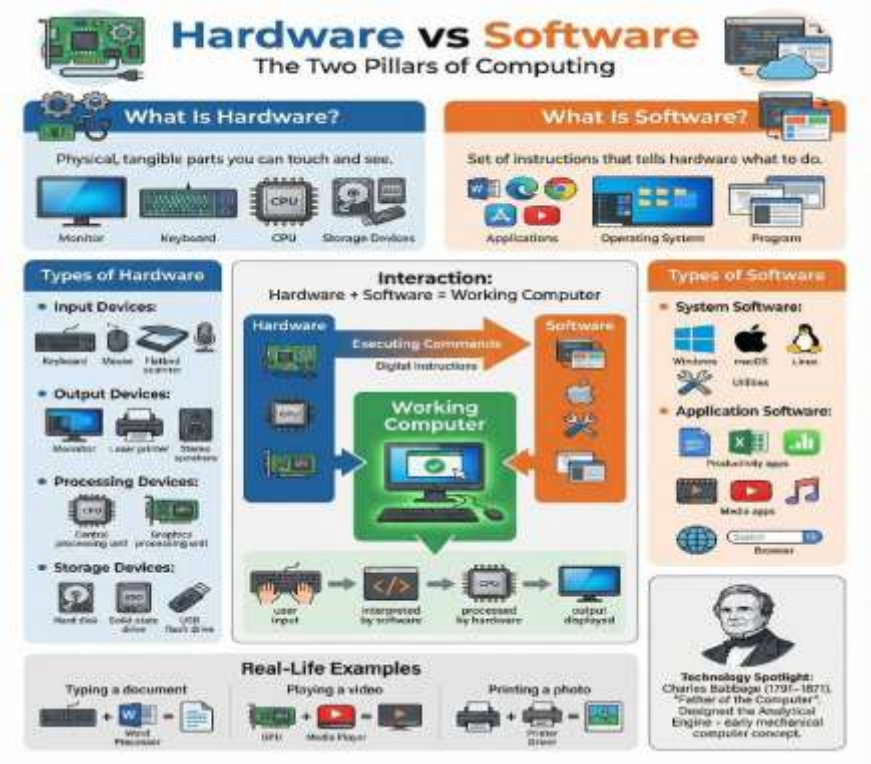
Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 3: Hardware & Software





Learning Objectives

By the end of this lesson, students will be able to:

- Define **hardware** and **software**.
- Identify common hardware devices.
- Identify different types of software.
- Explain how hardware and software work together.
- Differentiate between hardware and software.
- Perform simple practical activities using computer hardware and software.

Introduction

A computer works with the help of **Hardware** and **Software**.

- **Hardware** is the physical part of a computer that you can **see and touch**.
- **Software** is a set of **programs and instructions** that tell the computer what to do.

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Simple Example:

Think of a **human body**.

- 🧍 Body = Hardware
- 🧠 Brain & Instructions = Software

Both are needed to perform any task.

What is Hardware?

Hardware includes all the physical parts of a computer.

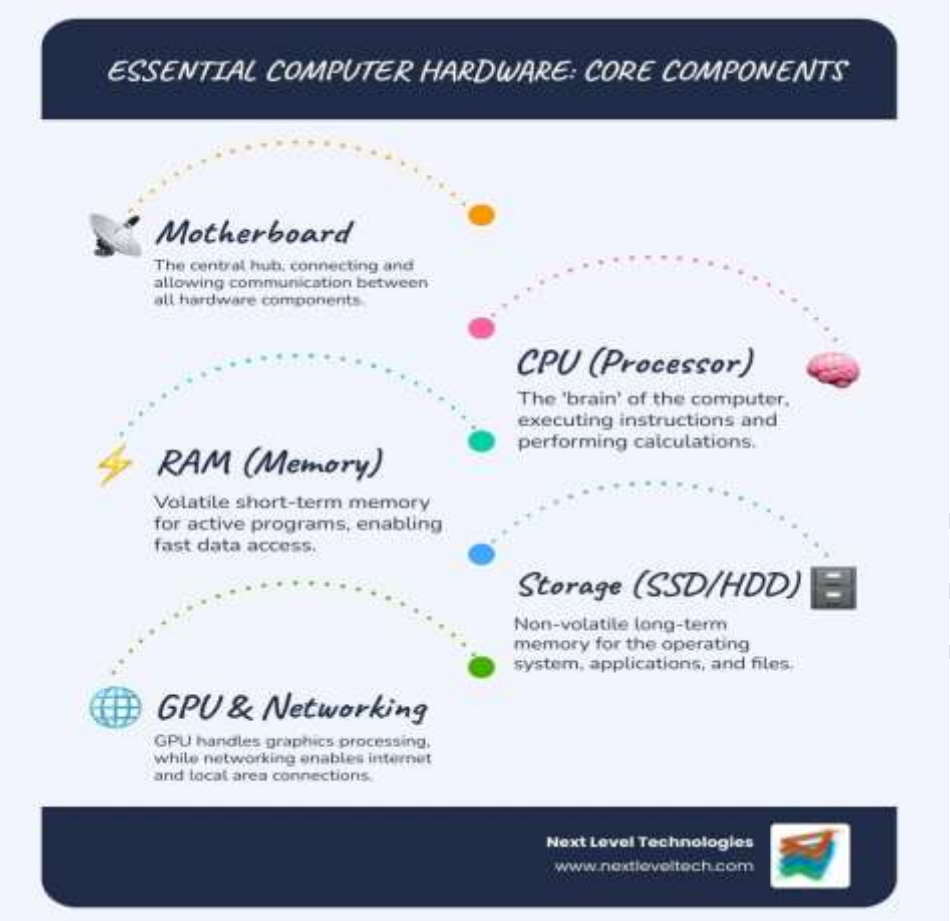
Definition

Hardware is the collection of physical components of a computer system that can be touched and seen.

Examples of Hardware



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Input Devices

Device	Function
Keyboard	Typing letters and numbers
Mouse	Clicking and selecting
Microphone	Recording voice
Webcam	Capturing images and videos
Scanner	Scanning documents

Output Devices

Device	Function
Monitor	Displays information
Printer	Prints documents
Speakers	Play sound
Headphones	Private listening

Storage Devices

Device	Function
Hard Disk (HDD)	Stores files and programs
SSD	Fast storage device
Pen Drive	Portable storage
CD/DVD	Stores music, videos, and files
Memory Card	Stores photos and videos

Processing Device

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Device	Function
🧠 CPU (Central Processing Unit)	Processes data and controls all operations

Characteristics of Hardware

- Physical components
- Can be touched
- Can wear out or break
- Needs software to work properly
- Can be repaired or replaced

What is Software?

Software is a collection of programs and instructions that tell the computer what to do.

Definition

Software is a set of programs that helps users perform different tasks on a computer.

Without software, hardware cannot perform useful work.

Types of Software

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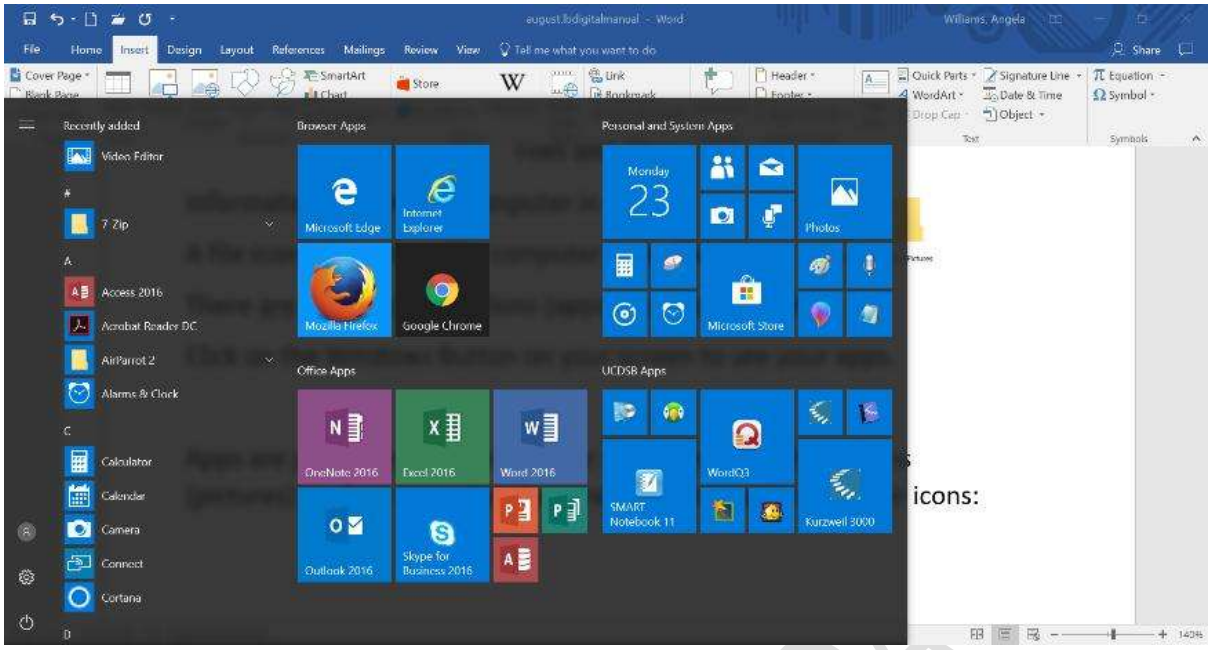
The Desktop



SYSTEM SOFTWARE
VERSUS
APPLICATION SOFTWARE

SYSTEM SOFTWARE	APPLICATION SOFTWARE
Computer software designed to provide a platform to other software	Software designed to perform a group of coordinated functions, tasks or activities for the benefit of the user
Manages resources and helps to run hardware and application software	Performs a specific task according to their type
Runs when the system starts and runs till the end	Runs when the user requires
Developed using languages like C, C++, Assembly	Developed using languages like Java, C, C++, Visual Basic
Essential for the proper functioning of a system	Not extremely important for the functioning of the system
Ex: operating system, language processors and device drivers	Ex: Word processor, Spreadsheet, Presentation software, web browsers, graphics software

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1. System Software

System software manages the computer and helps it run.

Examples

- Windows
- Linux
- macOS
- Android
- Device Drivers

Functions

- Starts the computer
- Manages memory
- Controls hardware
- Runs applications

2. Application Software

Application software helps users perform specific tasks.

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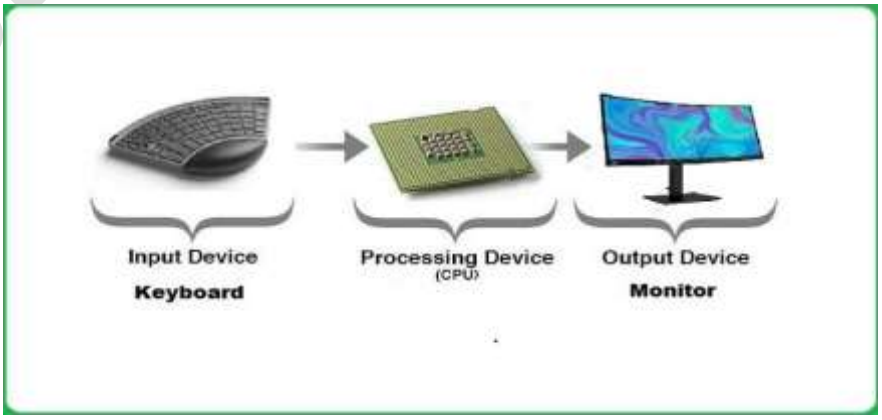
Examples

Software	Purpose
Microsoft Word	Typing documents
Microsoft PowerPoint	Creating presentations
MS Excel	Working with data
Paint	Drawing pictures
Calculator	Mathematical calculations
Google Chrome	Browsing the Internet
Python	Writing and running programs

Hardware vs Software

Hardware	Software
Physical parts of a computer	Programs and instructions
Can be touched	Cannot be touched
Manufactured in factories	Developed by programmers
May wear out or break	Can be updated or reinstalled
Examples: Keyboard, Monitor, CPU	Examples: Windows, Paint, Python

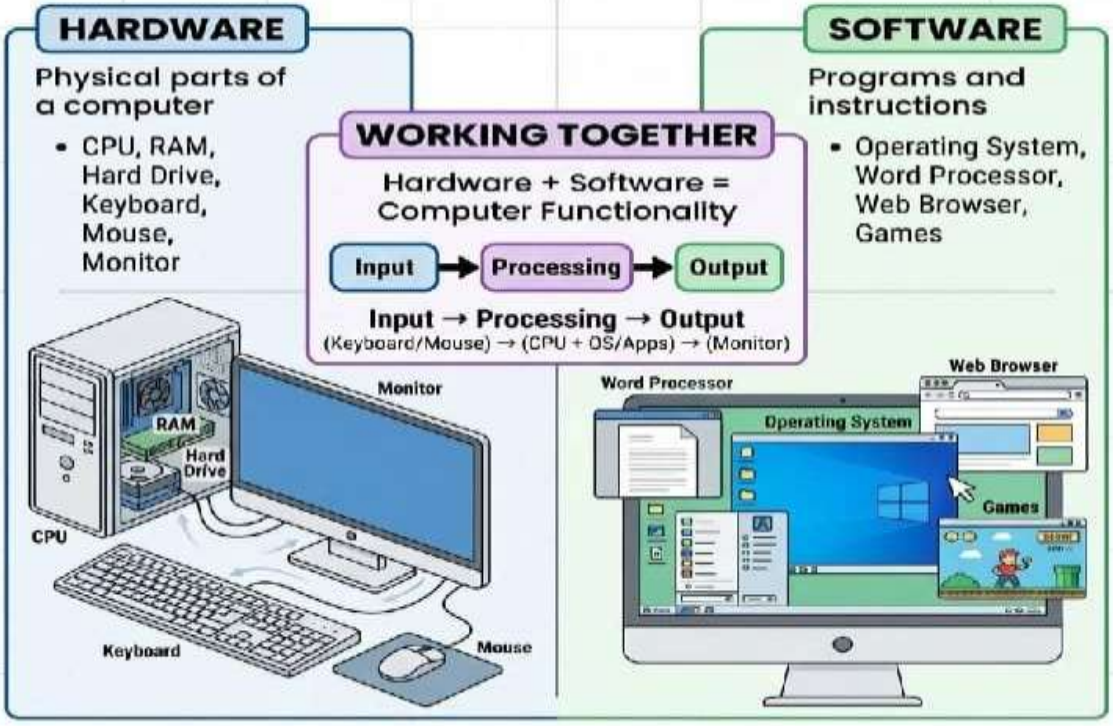
How Hardware and Software Work Together



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HARDWARE VS SOFTWARE

Understanding Computer Components



Input, process, output



Data is input.

The cpu decides what to do with it is output.

Example: Typing a Letter

1. You press a key on the **keyboard** (hardware).
2. The **operating system** (software) receives the input.
3. The **CPU** (hardware) processes the data.
4. **Microsoft Word** (software) displays the letters.
5. The **monitor** (hardware) shows the text.

Conclusion: Hardware and software must work together to complete every task.

Everyday Examples

Activity	Hardware	Software
Typing homework	Keyboard, Monitor	MS Word
Watching videos	Monitor, Speakers	Media Player
Drawing	Mouse, Monitor	Paint
Browsing websites	Keyboard, Mouse	Google Chrome
Learning Python	Keyboard, CPU	Python Interpreter

Advantages of Hardware

- Performs tasks quickly
- Stores large amounts of data
- Displays information clearly
- Makes communication easier

Advantages of Software

- Helps complete different tasks
- Easy to update
- Saves time
- Improves productivity

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- Makes learning fun and interactive

Computer Care Tips

- Keep the computer clean.
- Do not hit the keyboard or mouse.
- Shut down the computer properly.
- Avoid eating or drinking near the computer.
- Install software only from trusted sources.
- Keep antivirus software updated.

Practical Activities

Practical Activity 1 – Identify Hardware

Objective: Recognize computer hardware.

Instructions:

1. Observe a desktop or laptop.
2. Identify:
    - Monitor
    - Keyboard
    - Mouse
    - CPU/System Unit
    - Speakers
    - Printer (if available)
3. Write one function of each device.

Practical Activity 2 – Open Software

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Objective: Learn to use application software.

Instructions:

1. Turn on the computer.
2. Open **Paint**.
3. Draw a house using the mouse.
4. Save the drawing as **MyHouse.png**.
5. Close Paint.

Practical Activity 3 – Hardware or Software?

Write **H** for Hardware and **S** for Software.

Item	**Answer**
Monitor	H
Keyboard	H
Windows	S
Python	S
Printer	H
Google Chrome	S
Mouse	H
MS Paint	S
SSD	H
PowerPoint	S

Classroom Activity

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Hardware	Software Used
Keyboard	MS Word
Mouse	Paint
Monitor	Windows Display
Speakers	Media Player
Printer	Print Command

Discuss how each hardware device works with software.

Quick Quiz

1. Which of the following is hardware?

A) Windows

B) Keyboard ✅

C) MS Word

D) Paint

2. Which of the following is software?

A) Mouse

B) Printer

C) Google Chrome ✅

D) Monitor

3. Can hardware work without software?

A) Yes

B) No ✅

4. Which software is used for drawing?

A) Paint ✅

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B) Calculator

C) Browser

D) WordPad

5. Which device stores data permanently?

A) Keyboard

B) Hard Disk (HDD/SSD) ✅

C) Mouse

D) Monitor

Home Assignment

1. List **10 hardware devices** you have seen.
2. List **10 software applications** used at home or school.
3. Draw a computer and label at least **6 hardware parts**.
4. Write **5 differences** between hardware and software.
5. Ask your family members which software they use most often.

Lesson Summary

Today we learned:

- The meaning of **hardware** and **software**.
- Types of hardware: input, output, storage, and processing devices.
- Types of software: system software and application software.
- Differences between hardware and software.
- How hardware and software work together.
- Practical activities using computer hardware and software.

Learning Outcomes

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After completing this lesson, students will be able to:

- Define hardware and software.
- Identify common hardware components and software applications.
- Explain the difference between hardware and software.
- Describe how hardware and software work together to perform tasks.
- Use basic application software such as Paint or Word with confidence.
- Handle computer hardware safely and responsibly.

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